

Trainings ▾

Preparatory Steps

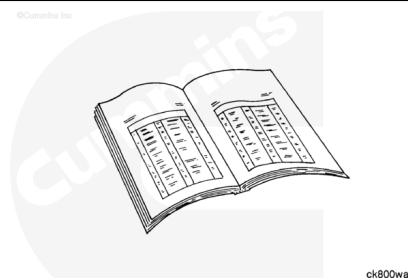
⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

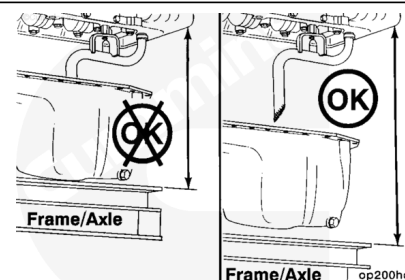
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/35/35-007-037-tr.html>)
- Remove the oil pan mounting capscrews. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/101/101-007-025-tr.html>)



Remove

Most front sump oil pans have a block-mounted suction tube, which requires approximately 610-mm [24-in] clearance between the vehicle frame or axle and the cylinder block oil pan mounting flange to remove the oil pan.

If the application will **not** allow this much clearance, lower the oil pan as much as possible, and reach inside the pan and remove the suction tube retaining capscrew from the cylinder block.

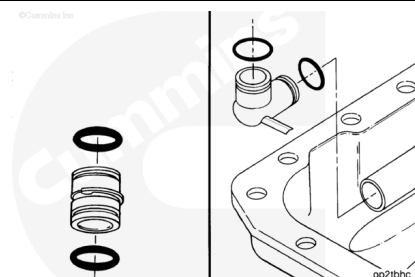


Let the suction tube lie inside the oil pan while the pan is removed from the chassis.

On rear sump pans, remove the oil transfer tube.

Remove and discard the o-ring(s) on the transfer tube or the suction tube.

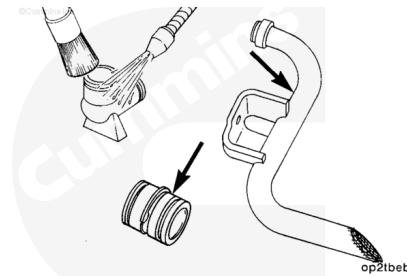
Note : The straight transfer tube is used on some aluminum front sump oil pan applications.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

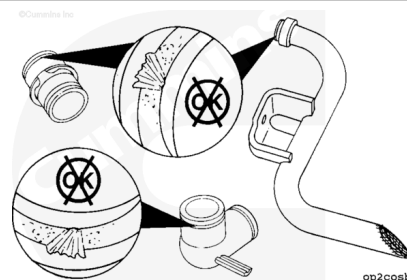
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent to clean the oil transfer tube and locating snap ring. Dry with compressed air.

Inspect the oil transfer tube for cracks or damage.

Inspect the o-ring grooves for cavitation or damage.

If cracks, damage, or cavitation is found, the oil transfer tube **must** be replaced.

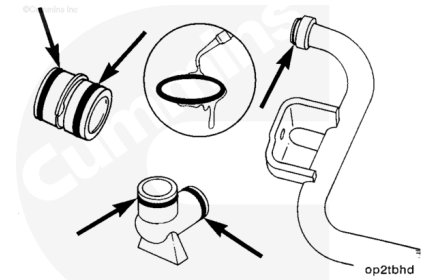


Install

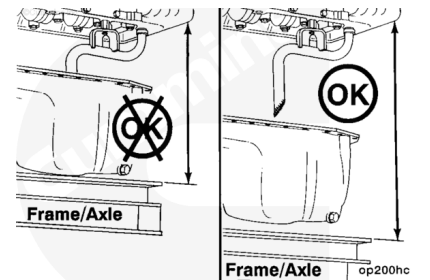
Install new o-ring(s) on the transfer tube or suction tube.

Use clean 15W-40 oil to lubricate the o-ring(s).

On rear sump oil pans, install the transfer tube in the oil pan.



On front sump oil pans, if the application does **not** allow approximately 610-mm [24-in] clearance between vehicle frame or axle and the cylinder block oil pan mounting surface during installation, the suction tube and oil pan will need to be installed together as an assembly. Installation requires approximately this much clearance to maneuver the oil pan over the frame or axle and under the suction tube when installing separately. Follow all the steps as outlined on the following pages.

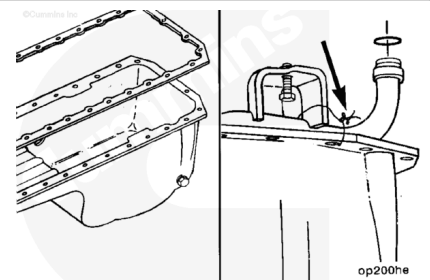


If the application does allow approximately 610-mm [24-in] clearance, install the suction tube first by itself, then the oil pan.

Install a new o-ring to the oil suction tube. Use clean 15W-40 engine oil to lubricate the o-ring.

Place a new gasket on the oil pan mounting flange.

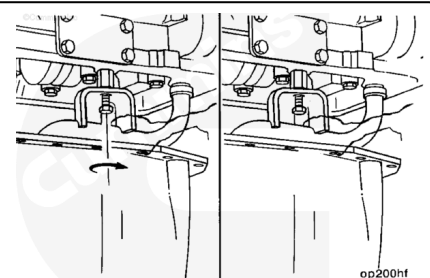
Attach the suction tube to the oil pan mounting flange with a piece of wire. This will hold the suction tube and oil pan together as one assembly until the tube is fastened to the cylinder block.



Maneuver the assembly into place under the engine in the chassis.

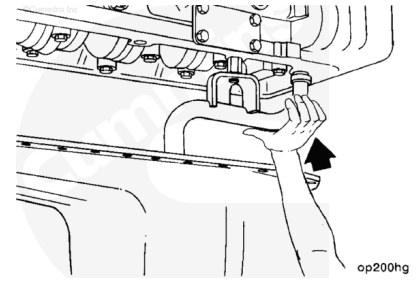
Align the suction tube with the cylinder block suction tube hole. Reach inside the engine, and start the capscrew into the mounting hole to hold the suction tube in place.

After the capscrew is started in the mounting hole by several threads, the wire holding the suction tube to the oil pan can be removed, and the oil pan can be lowered to allow more hand clearance to complete the suction tube installation.



⚠ CAUTION ⚠

Do not use the capscrew to pull the suction tube into place during installation. Doing so will cause misalignment of the suction tube which can cause damage to the engine.



Reach inside the engine, and push the suction tube completely up into place in the cylinder block suction tube mounting hole by hand.

Complete the capscrew installation.

Torque Value: 61 N·m [45 ft-lb]

Finishing Steps

⚠ CAUTION ⚠

Engine oil pressure must be indicated on the gauge within 15 seconds after starting. If oil pressure is not registered within 15 seconds, shut off the engine immediately to avoid engine damage. Confirm the correct oil level in the oil pan.



- Install the oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/101/101-007-025-tr.html>)
- Fill the engine with clean 15W-40 oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/35/35-007-037-tr.html>)
- Operate the engine at idle speed for 3 minutes to inspect for leaks.